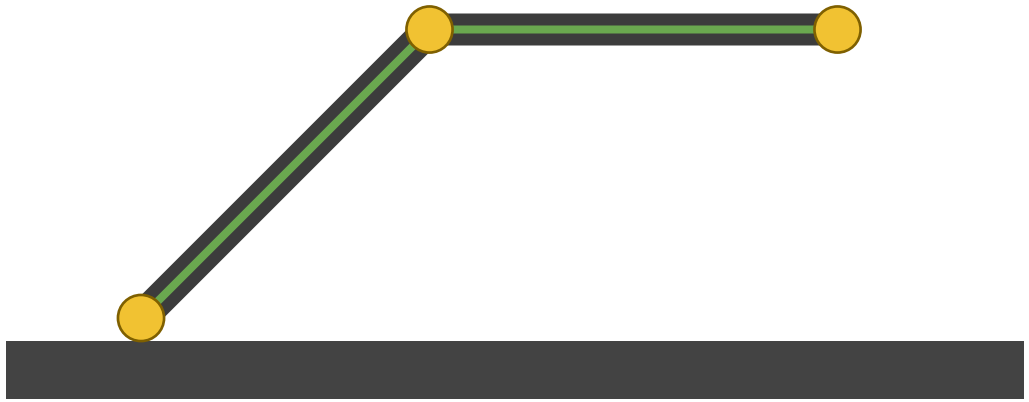
The DH Approach to Forward Kinematics

Robotic Arm Course · Lesson 4 of 10

Day 1: The flat arm with vectors
Day 2: Adding the base turn (3D)

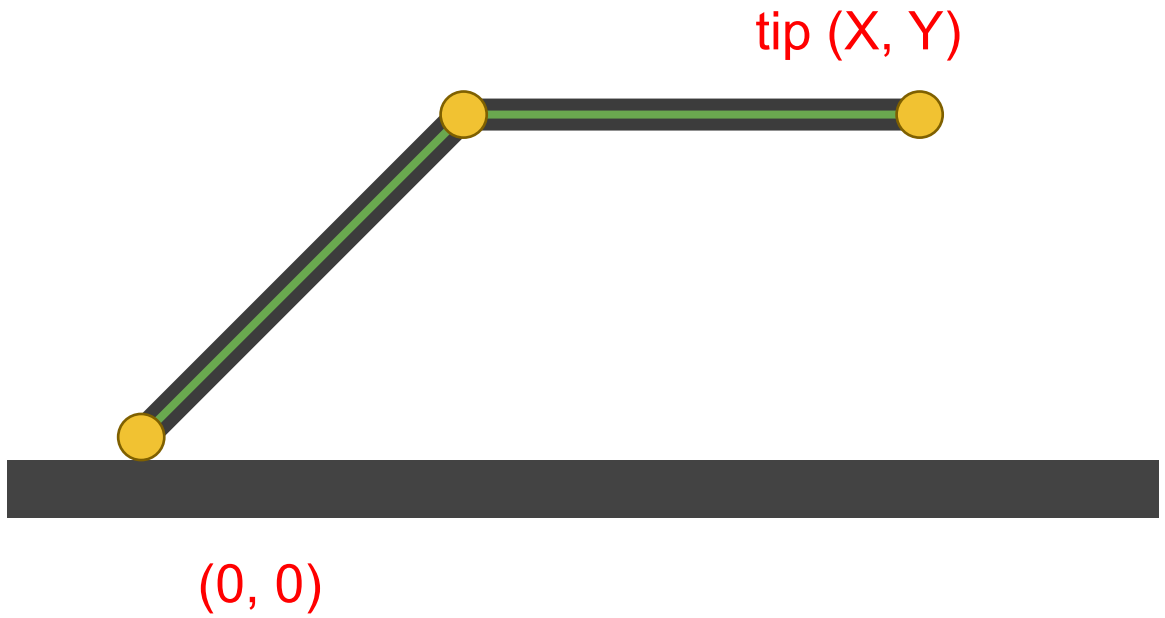


Where Lesson 3 left off:

L_1=250mm
L_2=250mm
θ_1=45°
θ_2=45°

In Lesson 3 we used triangles to find the tip of the arm: (X, Y) = (426.78, 176.78).

Today we get the same answer with vectors. We also measure the angles the way engineers do. This is called the DH convention.

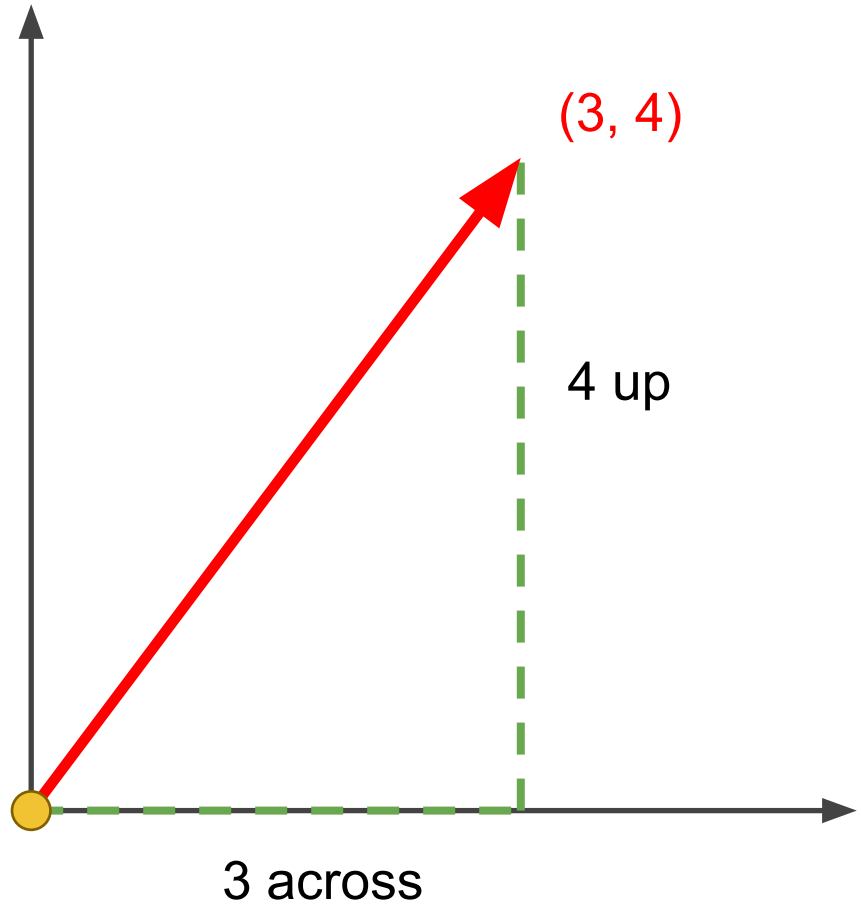


What is a vector?

A vector is an arrow.
It has a length and a direction.

We write a vector as numbers.
(3, 4) means 3 across and 4 up.

length = $\sqrt{(3^2 + 4^2)} = 5$

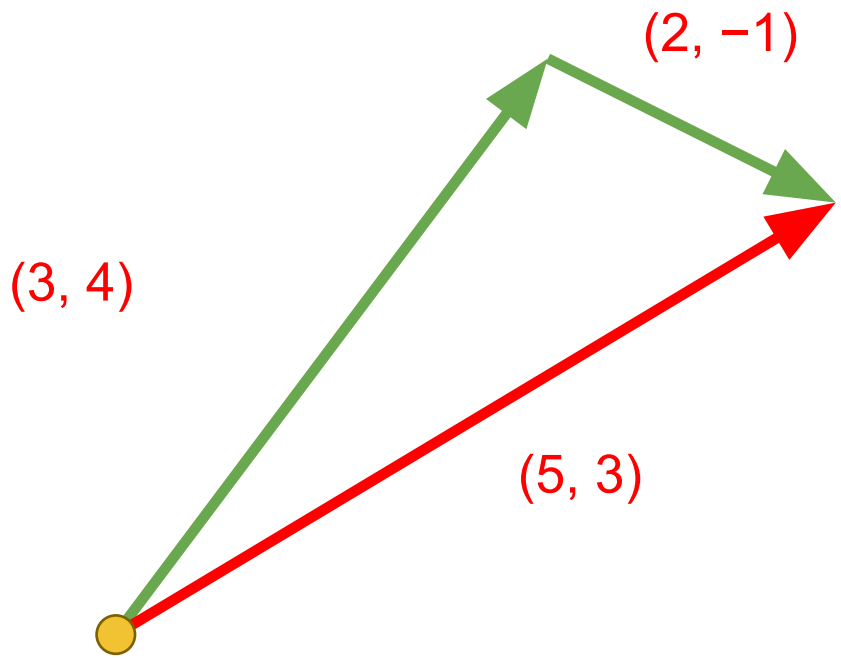


Adding vectors:

Walk (3, 4).
Then walk (2, −1).

You end up at the sum:
(3, 4) + (2, −1) = (5, 3)
Add the first numbers. Then add the second numbers.

Multiply by a number:
2 × (3, 4) = (6, 8)
Same direction, twice as long.

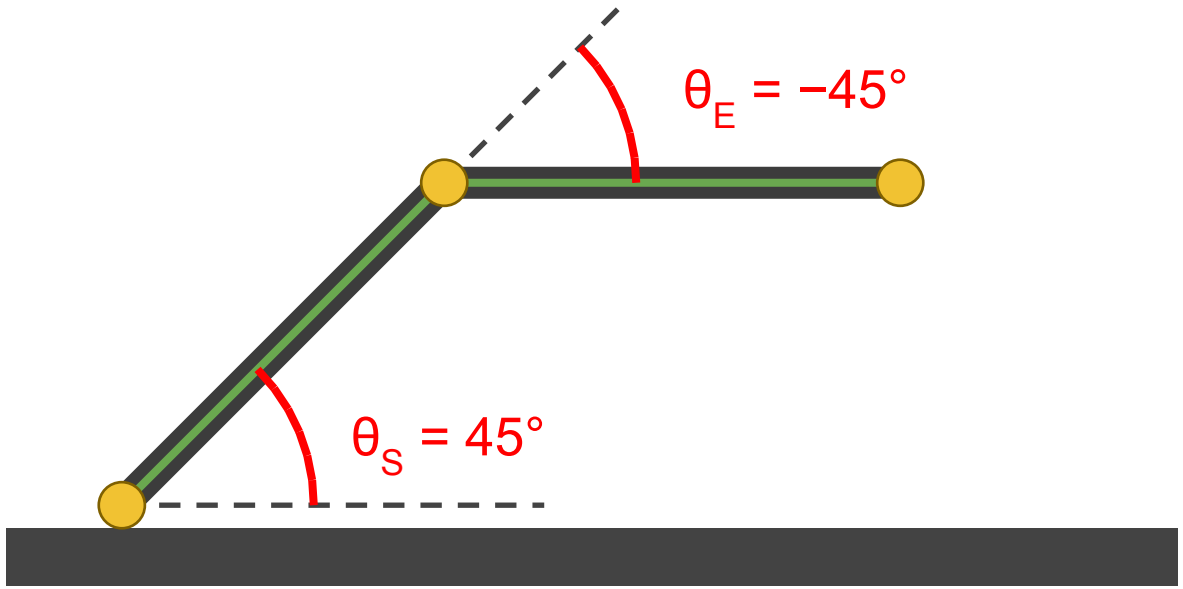


Measuring angles the DH way:

Each angle is measured from the link before it.
Up (counterclockwise) is +.
Down (clockwise) is −.

θ_S: the upper arm, measured from the floor.
θ_E: the forearm, measured from the line of the upper arm (dashed).

Our arm: $\theta_S = 45°$ and $\theta_E = -45°$



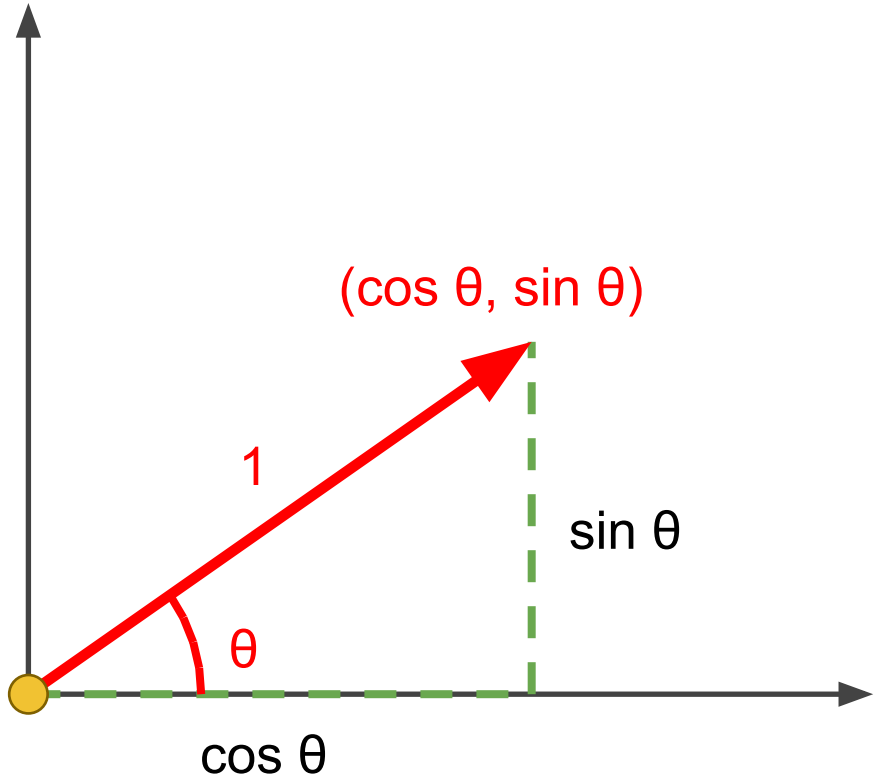
From Lesson 3: $\theta_S = 90° - \theta_1$, $\theta_E = -\theta_2$, $\theta_B = \theta_3$

An arrow for each link:

A unit vector is an arrow with length 1.
At angle θ from the floor, it is
(cos θ, sin θ)

Multiply it by the link length L to get the link vector.

Upper arm, $\theta_S = 45°$:
250 × (0.7071, 0.7071)
= (176.78, 176.78)

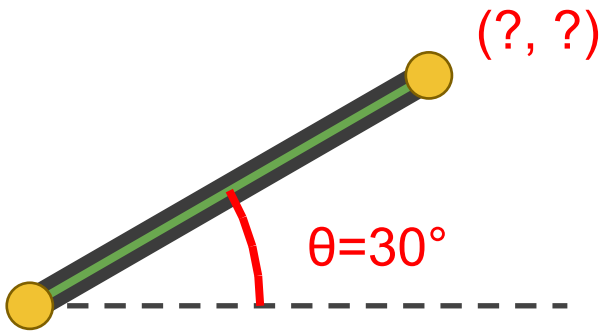


Try it yourself #1: find the link vector.

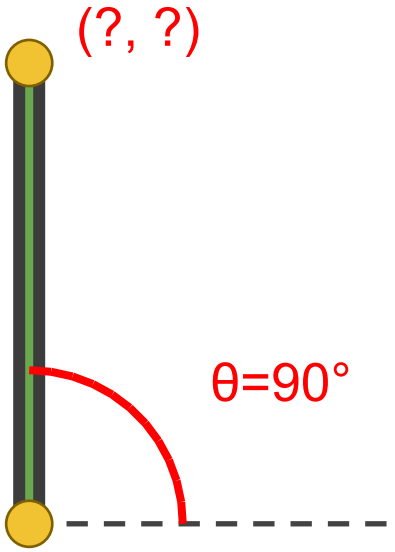
1. L=250mm
   θ=30°

2. L=250mm
   θ=90°

θ is measured from the floor.



1



2

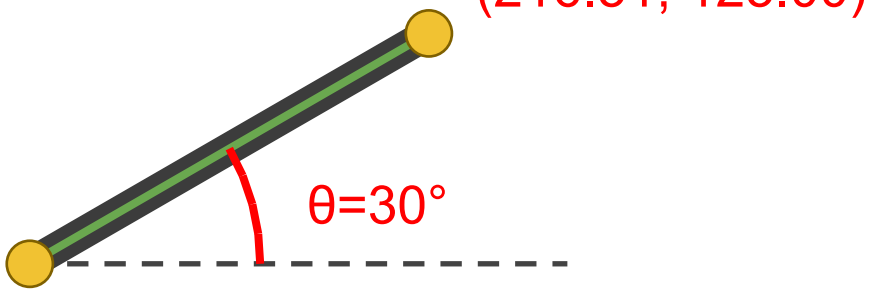
Try it yourself #1: solution

1. L=250mm
   θ=30°

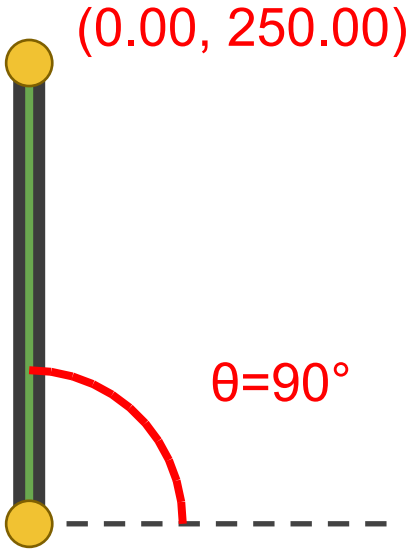
2. L=250mm
   θ=90°

Solution:

1. 250 × (cos 30°, sin 30°)
2. 250 × (cos 90°, sin 90°)



1



2

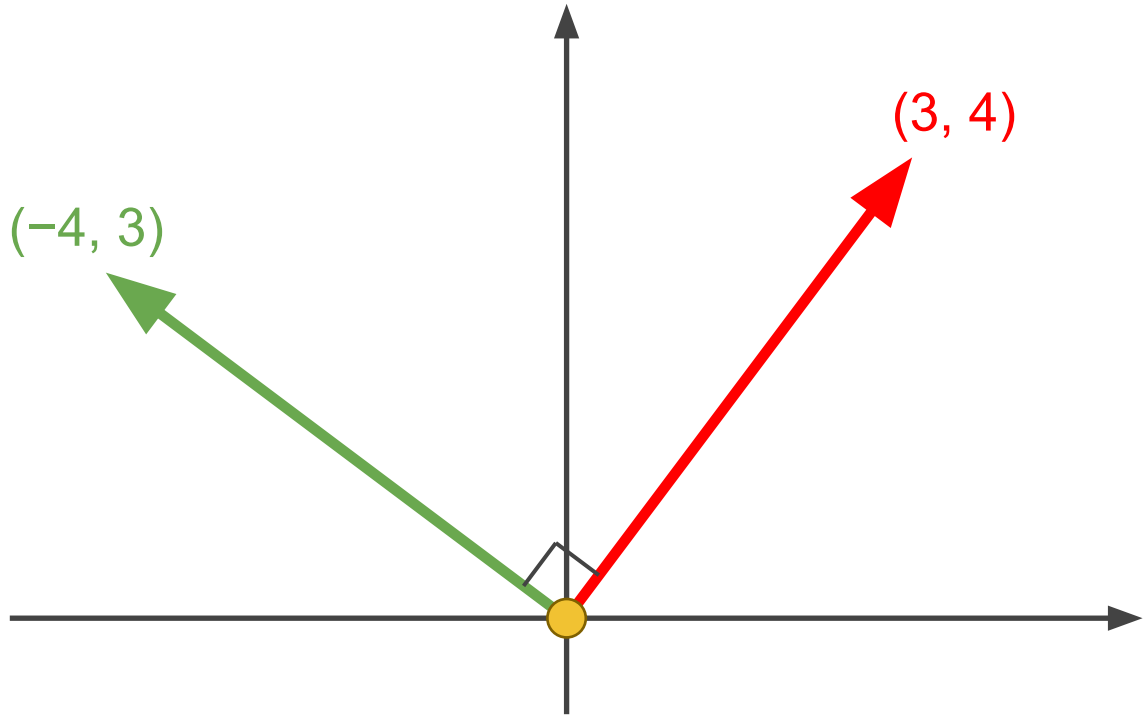
Turning an arrow 90°:

To turn an arrow 90° counterclockwise, swap the two numbers. Then change the sign of the first one.
(x, y) → (−y, x)

Example: (3, 4) → (−4, 3)

The upper arm: e = (0.7071, 0.7071)
turns into n = (−0.7071, 0.7071)

e and n are at right angles. Together they are the upper arm's frame.



Finding the forearm's arrow:

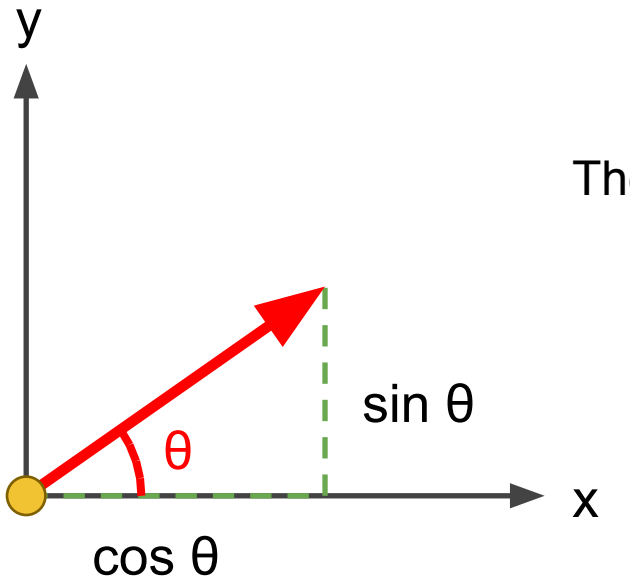
On the unit circle, an arrow at angle θ is:
cos θ along x, then sin θ along y.

The forearm's angle θ_E is measured from the upper arm, not from the floor. So at the elbow, tilt the axes:

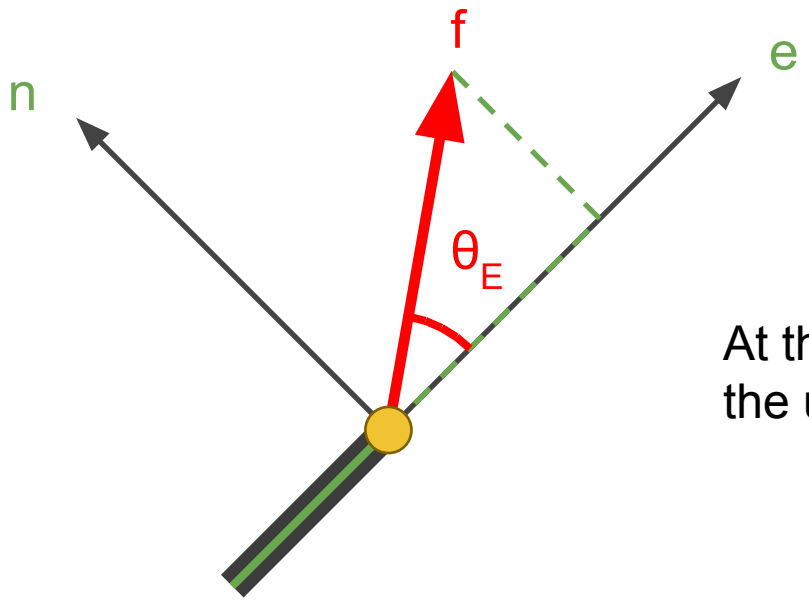
e (along the upper arm) is the new x
n (e turned 90°) is the new y

Use the same rule on the tilted axes:
$f = \cos\theta_E \cdot e + \sin\theta_E \cdot n$

cos θ_E along e, then sin θ_E along n.



The unit circle



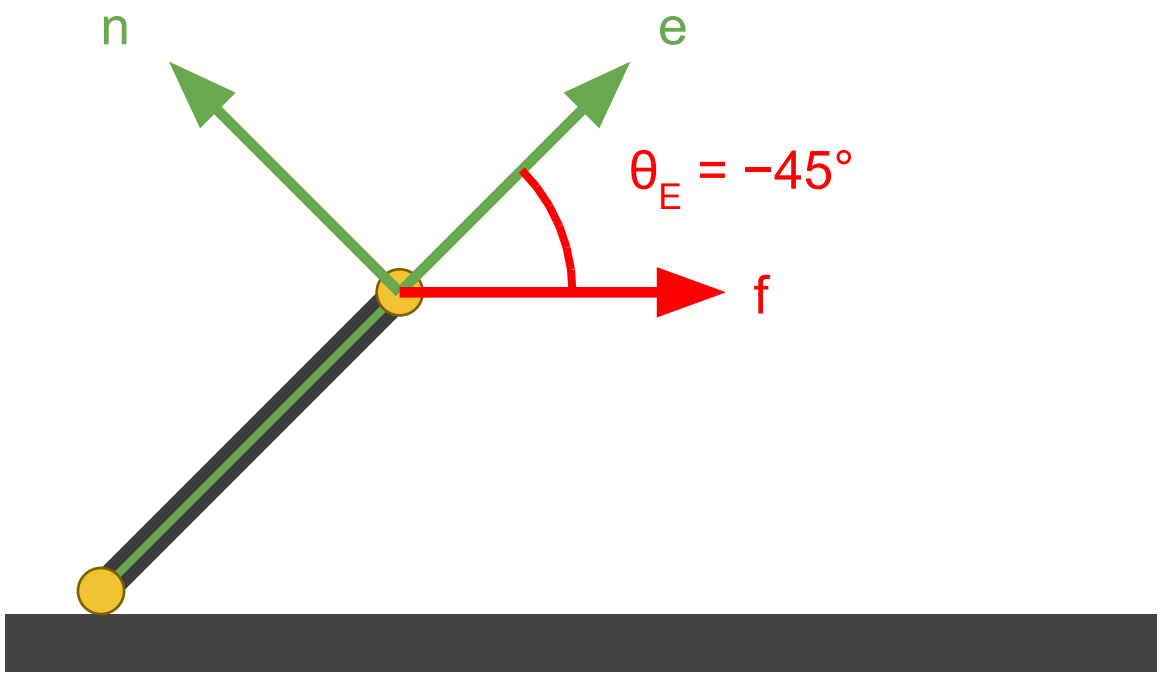
At the elbow: tilted with the upper arm

The forearm turns from the upper arm:

f is the forearm's arrow.
It is e turned by θ_E:
$f = \cos \theta_E \cdot e + \sin \theta_E \cdot n$

$\theta_E = 0°$: f = e (no turn)
$\theta_E = 90°$: f = n (a quarter turn up)
$\theta_E = -90°$: f = −n (a quarter turn down)

The forearm only needs the upper arm's frame. That is the DH idea.



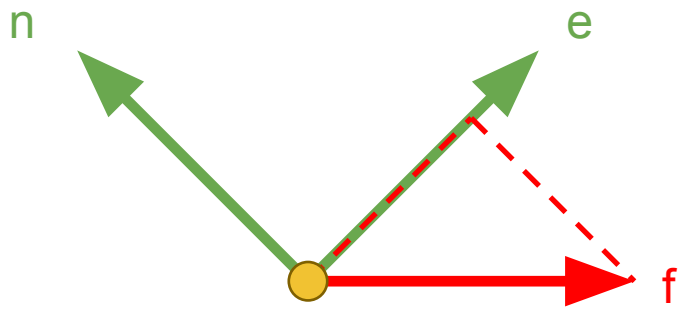
Worked example: the forearm's arrow

$\theta_S = 45°$ and $\theta_E = -45°$

e = (cos 45°, sin 45°) = (0.7071, 0.7071)
n = (−0.7071, 0.7071)
cos(−45°) = 0.7071 and sin(−45°) = −0.7071

f = 0.7071 × (0.7071, 0.7071) + (−0.7071) × (−0.7071, 0.7071)
  = (0.5000, 0.5000) + (0.5000, −0.5000)
  = (1.0000, 0.0000)

The forearm is flat. Check: 45° + (−45°) = 0° from the floor.



0.7071 along e, then 0.7071 away from n

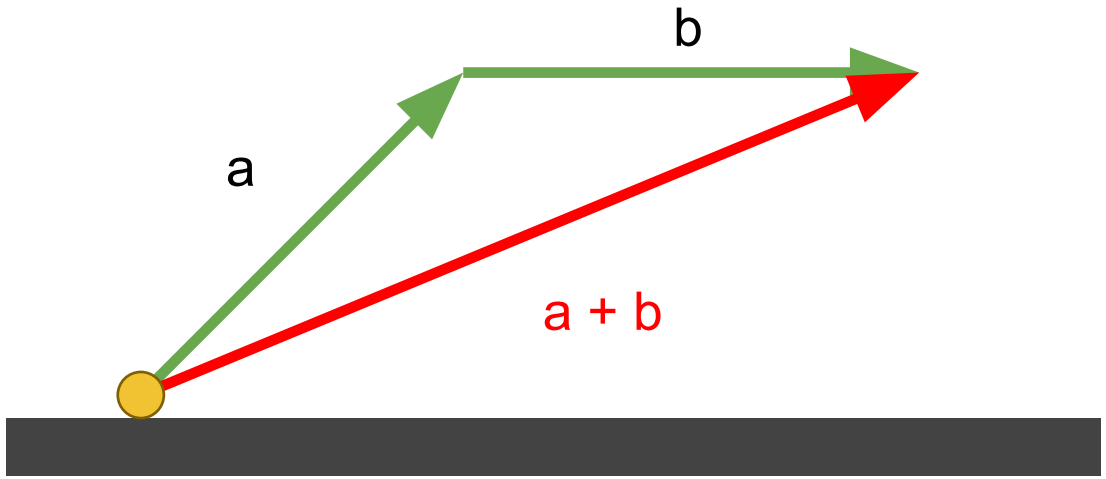
Two vectors, one tip:

Upper arm:
a = 250 × e = (176.78, 176.78)

Forearm:
b = 250 × f = (250.00, 0.00)

Put the arrows end to end:
a + b = (426.78, 176.78)

The same answer as Lesson 3.



The link table:

Link	**L (mm)**	**angle**	**arrow**	**vector = L × arrow**
Upper arm (e)	250	$\theta_S = 45°$	(0.7071, 0.7071)	(176.78, 176.78)
Turned 90° (n)			(−0.7071, 0.7071)	—
Forearm (f)	250	$\theta_E = -45°$	(1.0000, 0.0000)	(250.00, 0.00)
Tip (add)				(426.78, 176.78)

$e = (\cos \theta_S, \sin \theta_S)$ $n = (-\sin \theta_S, \cos \theta_S)$ $f = \cos \theta_E \cdot e + \sin \theta_E \cdot n$

Keep all the digits in your calculator. Round only at the end.

Try it yourself #2:

$L_1 = L_2 = 250\text{mm}$ $\theta_S = 60°$ $\theta_E = -30°$

Link	**L (mm)**	**angle**	**arrow**	**vector = L × arrow**
Upper arm (e)	250			
Turned 90° (n)				—
Forearm (f)	250			
Tip (add)				

(X, Y) = (?, ?)

Fill in the table, one row at a time.

In Lesson 3 this arm was $\theta_1 = 30°$ and $\theta_2 = 30°$.

Try it yourself #2: solution

$L_1 = L_2 = 250\text{mm}$ $\theta_S = 60°$ $\theta_E = -30°$

Link	**L (mm)**	**angle**	**arrow**	**vector = L × arrow**
Upper arm (e)	250	$\theta_S = 60°$	(0.5000, 0.8660)	(125.00, 216.51)
Turned 90° (n)			(−0.8660, 0.5000)	—
Forearm (f)	250	$\theta_E = -30°$	(0.8660, 0.5000)	(216.51, 125.00)
Tip (add)				(341.51, 341.51)

(X, Y) = (341.51, 341.51)

f = cos(−30°) e + sin(−30°) n
= 0.8660 e − 0.5000 n

In Lesson 3 this arm was $\theta_1 = 30°$ and $\theta_2 = 30°$.

Try it yourself #3:

$L_1 = L_2 = 250mm$ $\theta_S = 30°$ $\theta_E = 0°$

Link	**L (mm)**	**angle**	**arrow**	**vector = L × arrow**
Upper arm (e)	250			
Turned 90° (n)				—
Forearm (f)	250			
Tip (add)				

(X, Y) = (?, ?)

Fill in the table, one row at a time.
Use the rules on the link table slide.

In Lesson 3 this arm was $\theta_1 = 60°$ and $\theta_2 = 0°$.

Try it yourself #3: solution

$L_1 = L_2 = 250\text{mm}$ $\theta_S = 30°$ $\theta_E = 0°$

Link	L (mm)	angle	arrow	vector = L × arrow
Upper arm (e)	250	$\theta_S = 30°$	(0.8660, 0.5000)	(216.51, 125.00)
Turned 90° (n)			(−0.5000, 0.8660)	—
Forearm (f)	250	$\theta_E = 0°$	(0.8660, 0.5000)	(216.51, 125.00)
Tip (add)				(433.01, 250.00)

(X, Y) = (433.01, 250.00)

$\theta_E = 0°$, so f = e.
The arm is straight.

In Lesson 3 this arm was $\theta_1 = 60°$ and $\theta_2 = 0°$.

Day 1 recap:

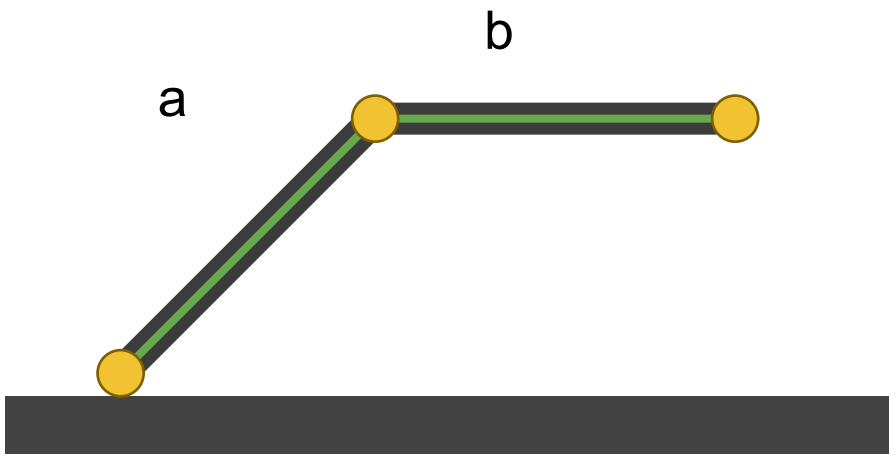
Each angle is measured from the link before it. Up is +. Down is −.

A link's arrow: $(\cos\theta, \sin\theta)$

Turn 90°: $(x, y) \to (-y, x)$

The forearm's arrow: $f = \cos\theta_E \cdot e + \sin\theta_E \cdot n$

The tip is the sum of the link vectors.



Day 2

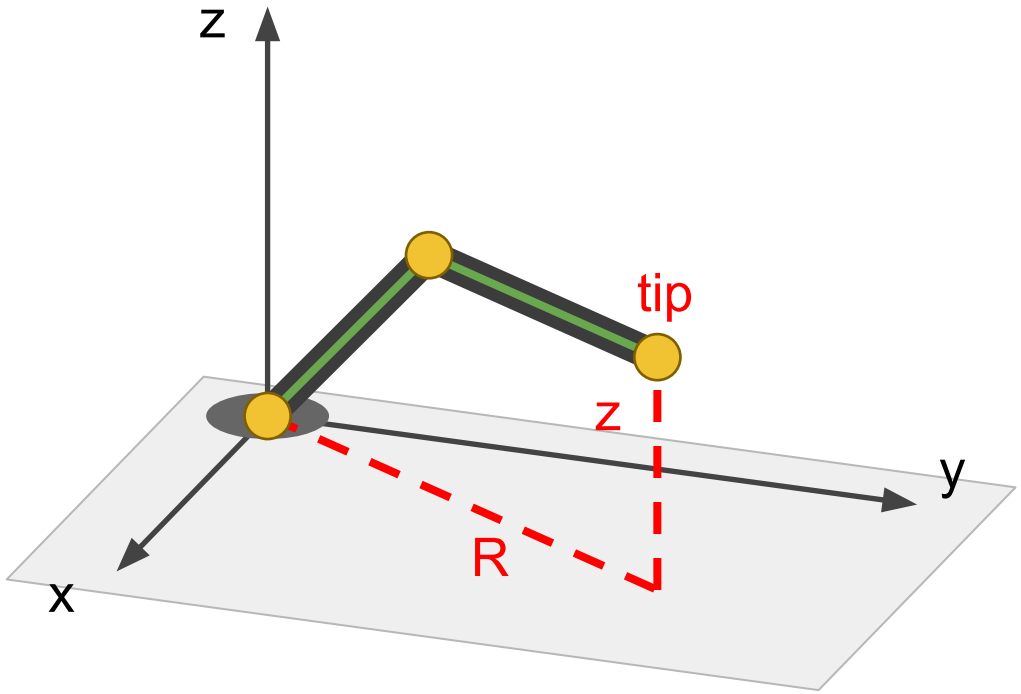
On Day 1 the tip was (across, up). For our arm it was (426.78, 176.78).

Today the base turns, so “across” can point any way on the floor. We give the two numbers new names:

the reach, R = 426.78 (along the floor)

the height, z = 176.78 (up)

From now on, up is z, like in Desmos. (Lesson 3 called it Y.)

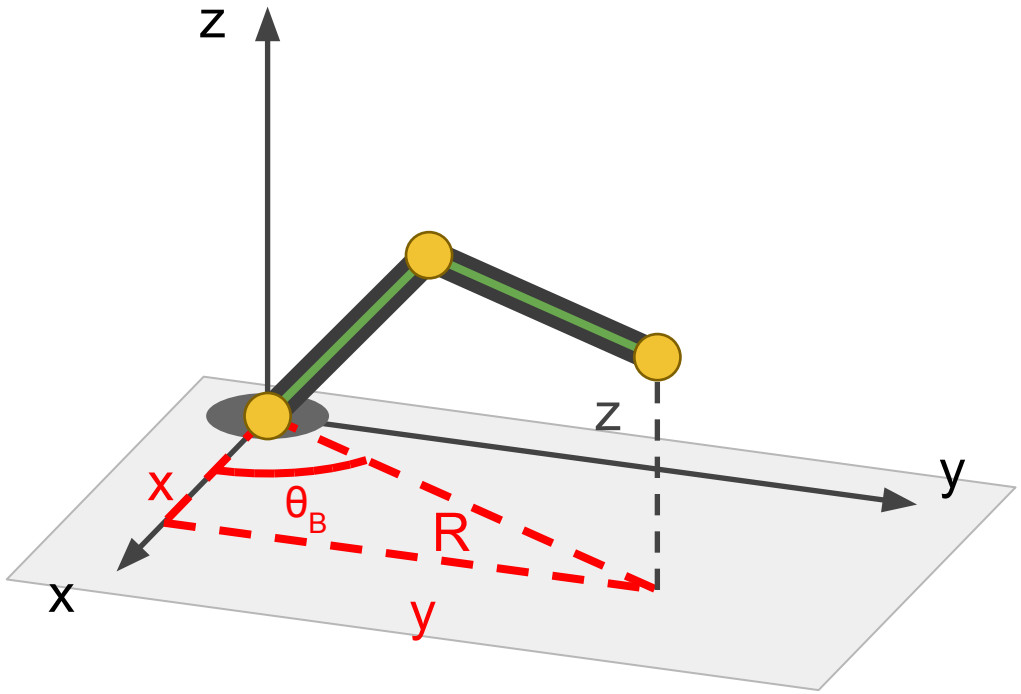


The base turns the whole arm:

The base turns the arm by θ_B.
The arm's shadow on the floor is the reach, R.

Split the shadow along x and y:
$x = R \cos \theta_B$
$y = R \sin \theta_B$

The height z does not change.



The upper arm's arrow in 3D:

Step 1. From the side, e looks like Day 1: length 1, at angle θ_S.

along the floor: $\cos \theta_S$

up: $\sin \theta_S$

Step 2. From above, the floor part points at angle θ_B. Split it like R on the last slide:

$x = \cos \theta_S \cos \theta_B$

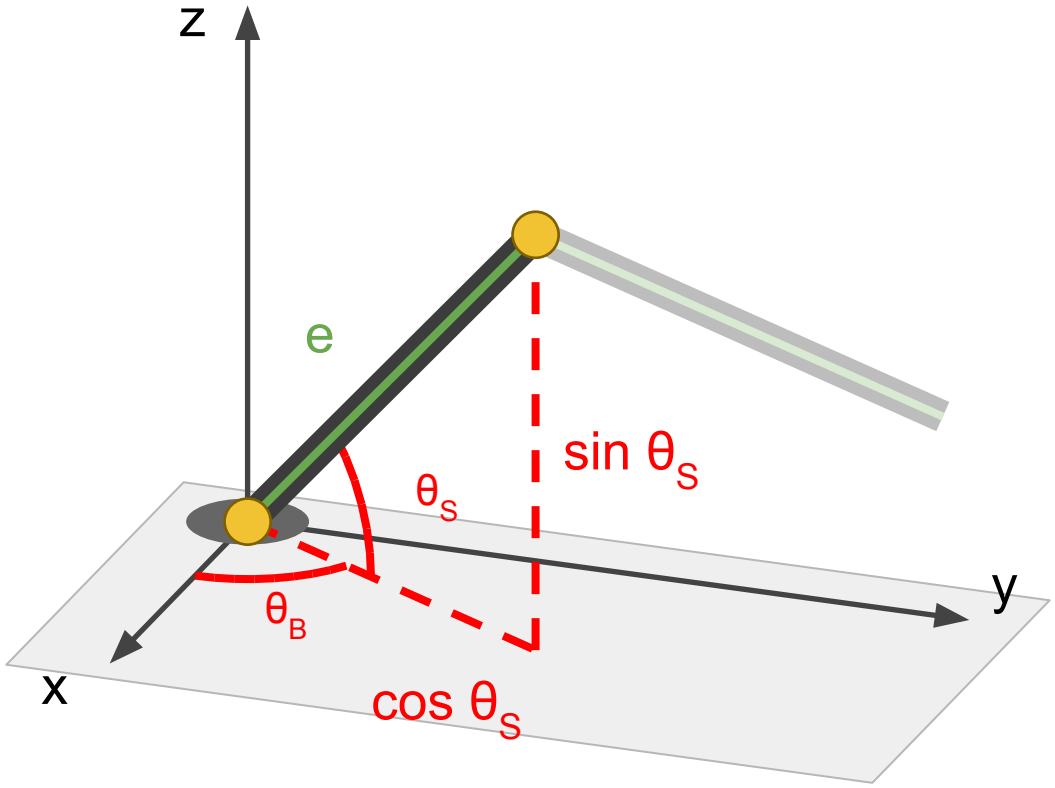
$y = \cos \theta_S \sin \theta_B$

Step 3. Turning the base does not change the height:

$z = \sin \theta_S$

So $e = (\cos \theta_S \cos \theta_B, \cos \theta_S \sin \theta_B, \sin \theta_S)$

In the Desmos graph, e is (a, b, c).



The turned arrow in 3D:

Step 1. From the side, turn e by 90° with the Day 1 rule $(x, y) \to (-y, x)$:

$(\cos \theta_S, \sin \theta_S) \to (-\sin \theta_S, \cos \theta_S)$

So n goes $-\sin \theta_S$ along the floor (back toward the base) and $\cos \theta_S$ up.

Step 2. From above, split the floor part with θ_B, the same as for e:

$x = -\sin \theta_S \cos \theta_B$

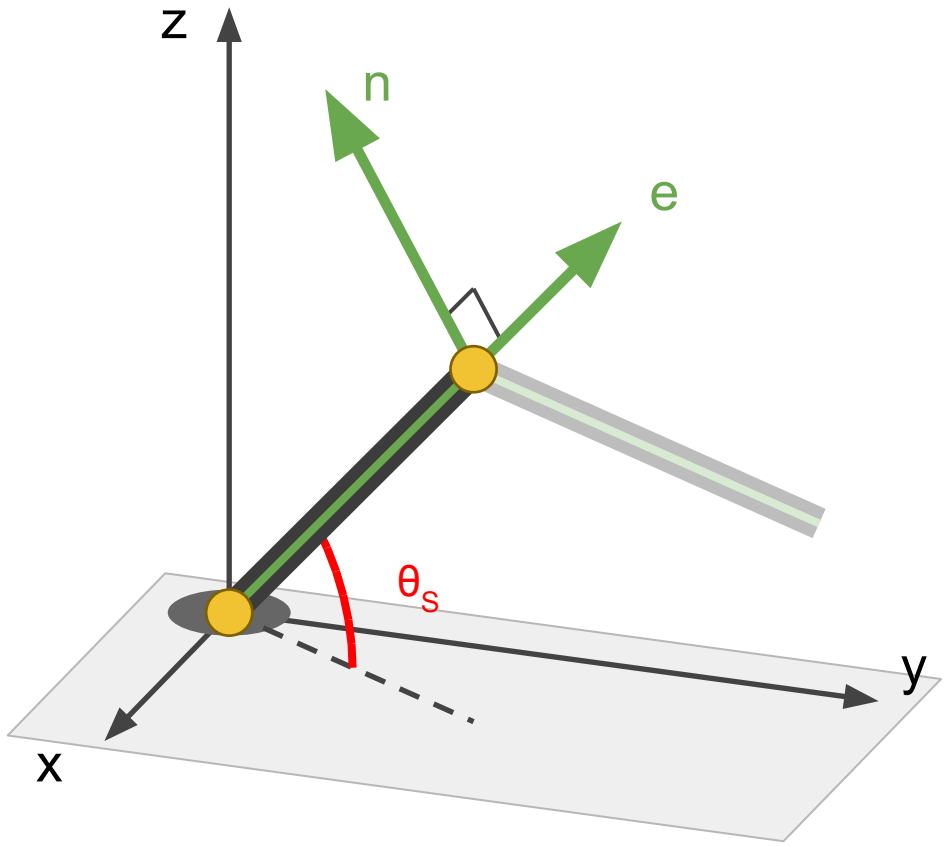
$y = -\sin \theta_S \sin \theta_B$

Step 3. The up part stays:

$z = \cos \theta_S$

So $n = (-\sin \theta_S \cos \theta_B, -\sin \theta_S \sin \theta_B, \cos \theta_S)$

Check: it is e with $\theta_S + 90°$ in place of θ_S.



e, n and the elbow's axle make the upper arm's frame.

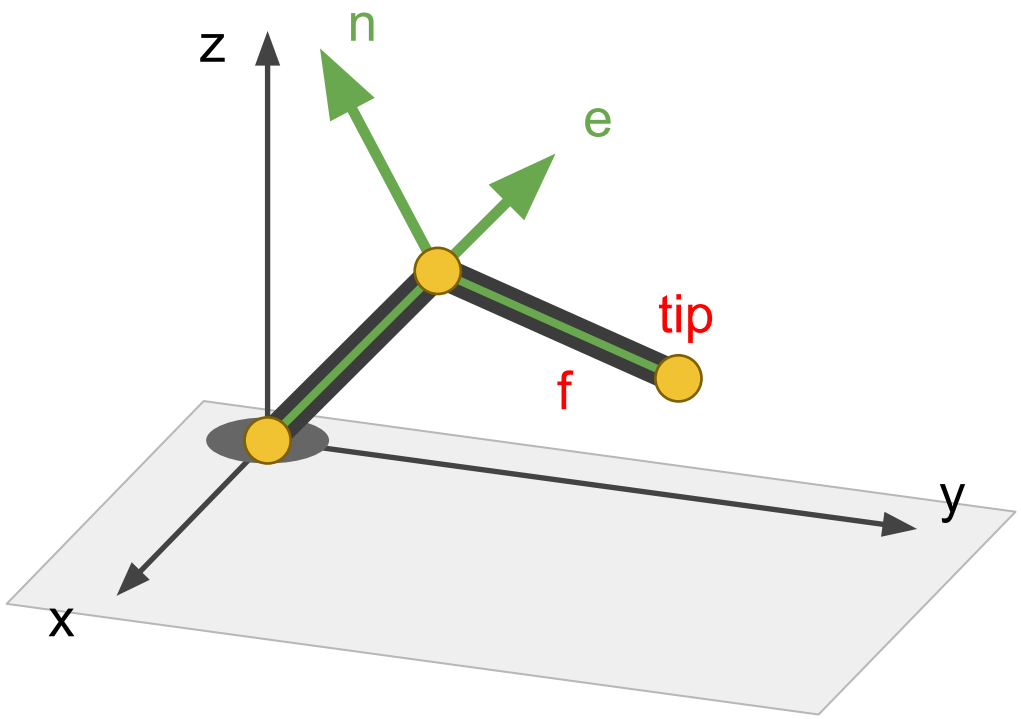
The forearm and the tip in 3D:

The same rules as Day 1:

$f = \cos \theta_E \cdot e + \sin \theta_E \cdot n$

$tip = L_1\, e + L_2\, f$

The base angle is already inside e and n. So we do not need a new rule.



Worked example: set up

$L_1 = L_2 = 250\text{mm}$ $\theta_B = 45°$ $\theta_S = 45°$ $\theta_E = -45°$

Link	L (mm)	angle	arrow (x, y, z)	vector (x, y, z)
Upper arm (e)	250	$\theta_S = 45°$	(0.5000, 0.5000, 0.7071)	(125.00, 125.00, 176.78)
Turned 90° (n)			(−0.5000, −0.5000, 0.7071)	—
Forearm (f)	250	$\theta_E = -45°$		
Tip (add)				

e = (cos 45° cos 45°, cos 45° sin 45°, sin 45°) = (0.5000, 0.5000, 0.7071)
n = (−sin 45° cos 45°, −sin 45° sin 45°, cos 45°) = (−0.5000, −0.5000, 0.7071)

Upper arm vector = 250 × e = (125.00, 125.00, 176.78)

Worked example: adding up

$L_1 = L_2 = 250\text{mm}$ $\theta_B = 45°$ $\theta_S = 45°$ $\theta_E = -45°$

Link	L (mm)	angle	arrow (x, y, z)	vector (x, y, z)
Upper arm (e)	250	$\theta_S = 45°$	(0.5000, 0.5000, 0.7071)	(125.00, 125.00, 176.78)
Turned 90° (n)			(−0.5000, −0.5000, 0.7071)	—
Forearm (f)	250	$\theta_E = -45°$	(0.7071, 0.7071, 0.0000)	(176.78, 176.78, 0.00)
Tip (add)				(301.78, 301.78, 176.78)

f = cos(−45°) e + sin(−45°) n = 0.7071 e − 0.7071 n
= (0.3536, 0.3536, 0.5000) + (0.3536, 0.3536, −0.5000) = (0.7071, 0.7071, 0.0000)

Forearm vector = 250 × f = (176.78, 176.78, 0.00)
Tip: add the vector column. (301.78, 301.78, 176.78)

Try it yourself #4:

$L_1 = L_2 = 250mm$ $\theta_B = 30°$ $\theta_S = 30°$ $\theta_E = 0°$

Link	**L (mm)**	**angle**	**arrow (x, y, z)**	**vector (x, y, z)**
Upper arm (e)	250			
Turned 90° (n)				—
Forearm (f)	250			
Tip (add)				

(x, y, z) = (?, ?, ?)

Fill in the table, one row at a time.

In Lesson 3 this arm was $\theta_1 = 60°$, $\theta_2 = 0°$, $\theta_3 = 30°$.

Try it yourself #4: solution

$L_1 = L_2 = 250mm$ $\theta_B = 30°$ $\theta_S = 30°$ $\theta_E = 0°$

Link	L (mm)	angle	arrow (x, y, z)	vector (x, y, z)
Upper arm (e)	250	$\theta_S = 30°$	(0.7500, 0.4330, 0.5000)	(187.50, 108.25, 125.00)
Turned 90° (n)			(−0.4330, −0.2500, 0.8660)	—
Forearm (f)	250	$\theta_E = 0°$	(0.7500, 0.4330, 0.5000)	(187.50, 108.25, 125.00)
Tip (add)				(375.00, 216.51, 250.00)

(x, y, z) = (375.00, 216.51, 250.00)

$\theta_E = 0°$, so f = e. The arm is straight.

In Lesson 3 this arm was $\theta_1 = 60°$, $\theta_2 = 0°$, $\theta_3 = 30°$.

Try it yourself #5:

$L_1 = L_2 = 250mm$ $\theta_B = 30°$ $\theta_S = 60°$ $\theta_E = -90°$

Link	**L (mm)**	**angle**	**arrow (x, y, z)**	**vector (x, y, z)**
Upper arm (e)	250			
Turned 90° (n)				—
Forearm (f)	250			
Tip (add)				

(x, y, z) = (?, ?, ?)

Fill in the table, one row at a time.

This is Practice Problem 2 from Lesson 3: $\theta_1 = 30°$, $\theta_2 = 90°$, $\theta_3 = 30°$.

Try it yourself #5: solution

$L_1 = L_2 = 250\text{mm}$ $\theta_B = 30°$ $\theta_S = 60°$ $\theta_E = -90°$

Link	L (mm)	angle	arrow (x, y, z)	vector (x, y, z)
Upper arm (e)	250	$\theta_S = 60°$	(0.4330, 0.2500, 0.8660)	(108.25, 62.50, 216.51)
Turned 90° (n)			(−0.7500, −0.4330, 0.5000)	—
Forearm (f)	250	$\theta_E = -90°$	(0.7500, 0.4330, −0.5000)	(187.50, 108.25, −125.00)
Tip (add)				(295.75, 170.75, 91.51)

(x, y, z) = (295.75, 170.75, 91.51)

$\theta_E = -90°$, so f = −n. The forearm points down: 60° − 90° = −30° from the floor.
Lesson 3 wrote the answer as (X, Y, Z) = (295.75, 91.51, 170.75), with Y up.

This is Practice Problem 2 from Lesson 3: $\theta_1 = 30°$, $\theta_2 = 90°$, $\theta_3 = 30°$.

Check it in Desmos:

1. Open the Desmos graph (link below).
2. Type the angles into u, v and w.
3. Read d, then multiply it by 100 to get mm.
   (The graph uses 2.5 for 250mm.)
4. Worked example: d = (3.0178, 3.0178, 1.7678)
   → (301.78, 301.78, 176.78)
5. Check Try it yourself #4 and #5.
6. To see the frames, click the gray circles
   in the Vectors folder.

$u = \theta_B$ (base)
$v = \theta_S$ (shoulder)
$w = \theta_E$ (elbow)

All in degrees.
v must be less than 90°.

https://www.desmos.com/3d/qq1w1zrs78

What the graph does:

In the graph	**In our steps**
$a = \cos(v)\cos(u),\ \ b = \cos(v)\sin(u),\ \ c = \sin(v)$	(a, b, c) is e, the upper arm's arrow
$(-ac, -bc, a^2 + b^2) \div \sqrt{(1 - c^2)}$	n, the turned arrow
$\cos(w)$ (a, b, c) + $\sin(w)$ × that arrow	f, the forearm's arrow
d = 2.5 × (forearm arrow) + 2.5 (a, b, c)	tip = L_1 e + L_2 f, in hundreds of mm
$(b, -a, 0) \div \sqrt{(1 - c^2)}$ (hidden)	the elbow's axle

Each link is one vector, and each joint is measured from the link before it.

What we learned:

Each joint angle is measured from the link before it.

A link's arrow is $(\cos\theta, \sin\theta)$. In 3D it is $(\cos\theta_S \cos\theta_B, \cos\theta_S \sin\theta_B, \sin\theta_S)$.

To turn an arrow by θ_E: $\cos\theta_E \cdot e + \sin\theta_E \cdot n$

The tip is the sum of the link vectors: $L_1\, e + L_2\, f$

The Desmos graph does the same steps.

Next lesson: we write these steps as a Python program.